Raghunandan Rajkumar

+46 72 76 57182 | raghunandan.officialuse@gmail.com |
LinkedIn: raghunandan-rajkumar | GitHub: rose-ember-forge
Portfolio: https://rose-ember-forge.github.io/

Physicist turned data scientist.
Passionate about turning complex datasets into actionable insight
through rigorous analysis and machine learning.

Experience

Research Scholar Jul 2018 – Nov 2021
Department of Physics, Pondicherry University *Pondicherry, India*

- Independently led a three-year research project using artificial neural networks and deep learning to model the functioning of biological neural networks.
- As a part of a research team, analyzed various EEG datasets and used Entropy measures to recognise human emotions, resulting in a published paper.
- **Publication:** *EEG-based Human Emotion Recognition using Entropy as a Feature Extraction Measure* `https://doi.org/10.1186/s40708-021-00141-5`.

Workshop Organizer 2019
Pondicherry University *Pondicherry, India*

- Conceived, planned, and led a technical workshop on LaTeX for physics students, managing everything from curriculum design to on-the-day facilitation.
- Translated complex typesetting concepts into accessible, hands-on content-making technical tools approachable for a non-specialist audience.
- Successfully promoted department-wide use of LaTeX as a better, streamlined method of academic typesetting.

Teacher 2017
Yuvaraja's College *Mysuru, India*

- Delivered master's-level physics courses, adapting content to diverse learning styles and fostering an environment where students felt comfortable engaging with challenging material.
- Mentored graduate students through complex problem-solving, building their confidence alongside their technical understanding.
- Delivered master's-level lectures in C-programming, connecting computational thinking to students' existing physics background to make abstract concepts concrete.

Technical Skills

Languages: Python, R, SQL, LaTeX, C
Data Science & Analysis: Data Visualization, NLP, Sentiment Analysis, Machine Learning, Deep Learning, Artificial Neural Networks
Tools & Libraries: Pandas, Scikit-Learn, PowerBI, Folium, Plotly, Spreadsheet software, Regular Expressions (RegEx)
Soft Skills: Problem Solving, Communication, Time Management, Leadership, Organization

Education

Master of Science in AI for Health Stockholm, Sweden
Stockholm University *2024 – 2026*

Master of Science (Physics) – CGPA: 6.86/10 Mysuru, India
Yuvaraja's College, University of Mysore *2016 – 2018*

Bachelor of Science (Physics, Chemistry, Mathematics) – 68.9% Mysuru, India
Yuvaraja's College, University of Mysore *2013 – 2016*

Projects

Why is my SL bus late? Stockholm public transport delays
Python,SQL, DuckDB, LightBGM, Streamlit

- Analysed 169 million stop events from one year of SL's real-time data. Buses rune on time 67% o fthe time against 96% for th emetro, and most bus delay builds up on the road (90$ on time at the first stop).
- MEasured the weather effect: each 10cm of snow on the group lowers punctuality by about 6 percentage points.
- Predicted arrivals 25% more accurately than SL's live predictions (67s v. 89s), and built a Streamlit dashboard with a map of delay hospots.

Skills in demand in Swedish data job ads, 2016 to 2026
SQL, DuckDB, Python, PowerBI, Streamlit

- Streamed and filtered 7.7 million Arbetsförmedlingen job ads down to 16-067 data jobs, with role classification, 44 skill patterns (239 of 240 audited matches correct) and repost removal.
- Found that Python rose from 18% to 37% of data-role ads and Azure from 5% to 21%, while SAS and R declined. Generative AI reached 41% of ML engineer ads by 2026, up from under 1% in 2022.
- Presented the results in a Power BI report (star schema, DAX measures) and a Streamlit dashboard.

Forecasting electricity prices in Stockholm
Python, LightGBM, scikit-learn, GitHub Actions

- Forecast tomorrow's 24 hourly electricity prices before the market closes, 44% more accurately than the standard benchmark over a full test year (average error 16.3 EUR/MWh).
- Added 80% prediction ranges that hold 80% of actual prices, so a user can plan around the uncertainty.
- Automated the pipeline with a GitHub Action that publishes a new forecast every morning, plus a Streamlit dashboard.

Certifications

Data Science for Engineers – NPTEL, IIT Madras (Oct 2022)

Data Science Professional Certificate – IBM via Coursera (Jun 2022)

Machine Learning – Stanford Online via Coursera (Aug 2021)

IELTS Academic – Overall Band Score: 8.0 (CEFR C1) (Oct 2022)

Languages

English: Native Proficiency (IELTS 8.0/C1 level)
Swedish: Advanced (B1/B2 level)
Kannada: Fluent
Malayalam: Fluent
Hindi: Advanced